# Limit, Continuity & differentiability for several variables.

Page No.: _____
Date: __/__/____

**Limit :** The function $f(x,y)$ is said to be the limit $l$ as $x \to a$ and $y \to b$ if and only if the limit $l$ is independent of the path followed by the point $(x,y)$ as $x \to a$, and $y \to b$ ;

$$\lim_{\substack{x \to a \\ y \to b}} f(x,y) = l$$

Note; Use simple equation ;

$y = mx + c$ : { to get relation between $x$ & $y$ }

① If $f(x,y) = f(m)$, then limit not exist.

② It apply only when $f(x,y) = \frac{0}{0}$ or $f(x,y) = \frac{\infty}{\infty}$

Akash

Q.1. $\lim_{\substack{x \to 1 \\ y \to 2}} \frac{2x^2y}{x^2+y^2+1}$

$\Rightarrow$ Given;

$$\lim_{\substack{x \to 1 \\ y \to 2}} \frac{2x^2y}{x^2+y^2+1}$$

$$\lim_{\substack{x \to 1 \\ y \to 2}} \frac{2(1)^2 \times 2}{(1)^2+(2)^2+1}$$

$$= \frac{4}{6}$$

$$= \frac{2}{3}$$

Q.2. $\lim_{\substack{x \to 0 \\ y \to 0}} \frac{xy}{x^2+y^2}$

$\Rightarrow$ Given;

$$\lim_{\substack{x \to 0 \\ y \to 0}} \frac{xy}{x^2+y^2} \qquad \left[\frac{0}{0} \text{ form}\right]$$

putting $y = mx + c$ at $x \to 0$, $y \to 0$

$$0 = 0 + c$$
$$c = 0$$

$$\therefore y = mx$$

Akash

$$\lim_{\substack{x \to 0 \\ mx \to 0}} \frac{x \cdot mx}{x^2 + m^2x^2}$$

$$\lim_{\substack{x \to 0 \\ mx \to 0}} \frac{mx^2}{x^2(1+m^2)}$$

$$\lim_{\substack{x \to 0 \\ mx \to 0}} \frac{m}{1+m^2}$$

$$\lim_{\substack{x \to a \\ y \to b}} f(x,y) = f(m)$$

$\therefore$ limit does n't exist

Q.3. $$\lim_{\substack{x \to 1 \\ y \to 1}} \frac{x(y-1)}{y(x-1)}$$

$\Rightarrow$ Given;

$$\lim_{\substack{x \to 1 \\ y \to 1}} \frac{x(y-1)}{y(x-1)} \qquad \left[\frac{0}{0} \text{ form}\right]$$

putting $y = mx + c$ at $x \to 1$, $y \to 1$

$1 = m + c$

$c = 1 - m$

$\therefore y = mx + 1 - m$

$y = m(x-1) + 1$

Akash

$$\cancel{\lim_{\substack{x\to 1\\ m(x+1)+m\to 1}} \frac{x(m(x+1)+1-1)}{(m(x+1)+1)(x-1)}}$$

$$\cancel{\lim_{\substack{x\to 1\\ m(x+1)+m\to 1}} \frac{mx(x+1)}{(x-1)[m(x+1)+1)]}}$$

$$\Rightarrow \lim_{\substack{x\to 1\\ m(x-1)+1\to 1}} \frac{x[m(x-1)+1-1]}{[m(x-1)+1](x-1)}$$

$$\Rightarrow \lim_{\substack{x\to 1\\ m(x-1)+1\to 1}} \frac{mx\cancel{[x-1]}}{[m(x-1)+1]\cancel{(x-1)}}$$

$$\Rightarrow \lim_{\substack{x\to 1\\ m(x-1)+1\to 1}} \frac{mx}{[m[x-1]+1]}$$

$$\lim_{\substack{x\to 1\\ m(x-1)+1\to 1}} m$$

$$\lim_{\substack{x\to a\\ y\to b}} f(x,y) = m.$$

Limit doesn't exist.

Akash

Page No.:
Date: / /

→ Continuity :- A function $f(x,y)$ is said to be continuous at the point $(a,b)$ if

$$\lim_{\substack{x \to a \\ y \to b}} f(x,y) \text{ exist and} = f(a,b)$$

i.e. $\lim_{\substack{x \to a \\ y \to b}} f(x,y)$ exist it means $\neq$ to $f(m)$ and equal to $f(a,b)$ that will given in the question.

PYQ-2022

Q.1. Show the function $f(x,y) = \begin{cases} \dfrac{xy}{\sqrt{x^2+y^2}} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$

is continuous at origin.

→ Given;

$$f(x,y) = \begin{cases} \dfrac{xy}{\sqrt{x^2+y^2}} & , (x,y) \neq 0 \\ 0 & , (x,y) = (0,0) \end{cases}$$

Now,

to check continuous at origin i.e. $(x,y) = (0,0)$

putting $y = mx + c$ at $x = 0$, $y = 0$

$0 = 0 + c$

$c = 0$

$\therefore y = mx$

Akash

$$f(x,y) = \frac{x \cdot mx}{\sqrt{x^2+m^2x^2}}$$

$$f(x,mx) = \frac{mx^2}{\sqrt{x^2[1+m^2]}}$$

$$f(x,mx) = \frac{mx^2}{x\sqrt{1+m^2}}$$

$$f(x,mx) = \frac{mx}{\sqrt{1+m^2}}$$

$$\lim_{\substack{x\to 0\\ mx\to 0}} f(x,mx) = \lim_{\substack{x\to 0\\ mx\to 0}} \frac{mx}{\sqrt{1+m^2}}$$

$$\lim_{\substack{x\to 0\\ mx\to 0}} f(x,mx) = 0$$

$$\lim_{\substack{x\to 0\\ y\to 0}} f(x,y) = 0 = f(0,0)$$

Hence, it is continuous at origin.

Q.2. Discuss the continuity of the following function at the point (0,0)

$$f(x,y) = \begin{cases} \dfrac{x^2y^2}{x^3+y^3} & \text{for } (x,y) \neq (0,0) \\ 0 & , f(x,y) = (0,0) \end{cases}$$

Akash

Page No.:
Date: __/__/__

⇒ Given;

$$f(x,y) = \frac{x^2y^2}{x^3+y^3}$$

to check continuity at point (0,0).

putting $y = mx + c$ at $x \to 0$, $y \to 0$

$0 = 0 + c$

$c = 0$

$\therefore y = mx$

$$f(x,y) = \frac{x^2 \cdot m^2 x^2}{x^3 + m^3 x^3}$$

$$f(x,mx) = \frac{x^4 [m^2]}{x^3 [1+m^3]}$$

$$f(x,mx) = \frac{x m^2}{1+m^3}$$

$$\underset{\substack{x \to 0 \\ mx \to 0}}{Lt} f(x,mx) = \underset{\substack{x \to 0 \\ mx \to 0}}{Lt} \frac{xm^2}{1+m^3}$$

$$\underset{\substack{x \to 0 \\ y \to 0}}{Lt} f(x,y) = 0 = f(0,0)$$

Hence, it is continuous function at the point (0,0).

Akash

Date: __/__/__

Q. 3. $f(x,y) = \begin{cases} \dfrac{x^2-y^2}{x^2+y^2} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$

Check continuity at (0,0).

=> Given;

$$f(x,y) = \frac{x^2-y^2}{x^2+y^2}$$

to check continuity at (0,0)

putting $y = mx + c$ at $x \to 0$, $y \to 0$

$$0 = 0 + c$$
$$c = 0$$
$$\therefore y = mx$$

$$f(x,y) = \frac{x^2 - m^2x^2}{x^2 + m^2x^2}$$

$$f(x, mx) = \frac{\cancel{x^2}[1-m^2]}{\cancel{x^2}[1+m^2]}$$

$$f(x, mx) = \frac{1-m^2}{1+m^2}$$

$$f(x,y) = f(m)$$

Hence, limit doesn't exist

$\therefore$ function is not continuous.

Akash

Page No.:
Date: / /

MB-2012

Q.4. Show that the following function is continous at the point (0,0):

$$f(x,y) = \begin{cases} \dfrac{2x^4+3y^4}{x^2+y^2} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$$

→ Given;

$$f(x,y) = \frac{2x^4+3y^4}{x^2+y^2}$$

Now, to check continuity at (0,0)

putting $y = mx + c$ at $x \to 0$, $y \to 0$

$$0 = 0 + c$$
$$c = 0$$
$$\therefore y = mx$$

$$f(x,y) = \frac{2x^4 + 3m^4x^4}{x^2 + m^2x^2}$$

$$f(x,mx) = \frac{x^4[2+3m^4]}{x^2[1+m^2]}$$

$$f(x,mx) = \frac{x^2[2+3m^4]}{1+m^2}$$

$$\lim_{\substack{x\to 0 \\ mx \to 0}} f(x,mx) = \lim_{\substack{x\to 0 \\ mx \to 0}} \frac{x^2[2+3m^4]}{1+m^2}$$

Akash

$$\lim_{\substack{x \to 0 \\ mx \to 0}} f(x, mx) = 0$$

$$\lim_{\substack{x \to 0 \\ y \to 0}} f(x,y) = 0 = f(0,0)$$

Hence, the function is continuous.

PYQ-2025

Q.5. Discuss the continuity of the function $f(x,y)$ at the given point

$$f(x,y) = \begin{cases} \dfrac{x^2+y^2}{\tan xy} & , (x,y) \neq (0,0) \\ 0 & , (x,y) = (0,0) \end{cases}$$

$\Rightarrow$ Given;

$$f(x,y) = \frac{x^2+y^2}{\tan xy}$$

Now, to check continuity at $(0,0)$

$$\lim_{\substack{x \to 0 \\ y \to 0}} \frac{x^2+y^2}{\tan xy}$$ [multiply & divide $xy$]

Akash

$$\Rightarrow \lim_{\substack{x\to 0\\ y\to 0}} \frac{x^2+y^2}{\tan xy} \cdot \frac{xy}{xy}$$

$$\Rightarrow \lim_{\substack{x\to 0\\ y\to 0}} \frac{x^2+y^2}{xy} \cdot \lim_{\substack{x\to 0\\ y\to 0}} \frac{xy}{\tan xy}$$

$$\Rightarrow \lim_{\substack{x\to 0\\ y\to 0}} \frac{x^2+y^2}{xy} \cdot \underbrace{\lim_{\substack{x\to 0\\ y\to 0}} \frac{1}{\frac{\tan xy}{xy}}}_{1 \left\{\lim_{x\to 0} \frac{\tan x}{x} = 1\right\}}$$

$$\rightarrow \lim_{\substack{x\to 0\\ y\to 0}} \frac{x^2+y^2}{xy}$$

putting $y = mx + c$ at $x \to 0$ & $y \to 0$

$0 = 0 + c,$

$c = 0$

$\therefore y = mx$

$$\lim_{\substack{x\to 0\\ mx\to 0}} \frac{x^2+m^2x^2}{x \cdot mx}$$

$$\lim_{\substack{x\to 0\\ mx\to 0}} \frac{\cancel{x^2}(1+m^2)}{\cancel{x^2}m}$$

Akash

$$\lim_{\substack{x \to 0 \\ mx \to 0}} \frac{1+m^2}{m}$$

$$\lim_{\substack{x \to 0 \\ y \to 0}} f(x,y) = f(m)$$

Hence, limit doesn't exist

$\therefore$ function is not continous.

Akash